

# Lessons Learned from 8 Years of Open OnDemand at Princeton Research Computing

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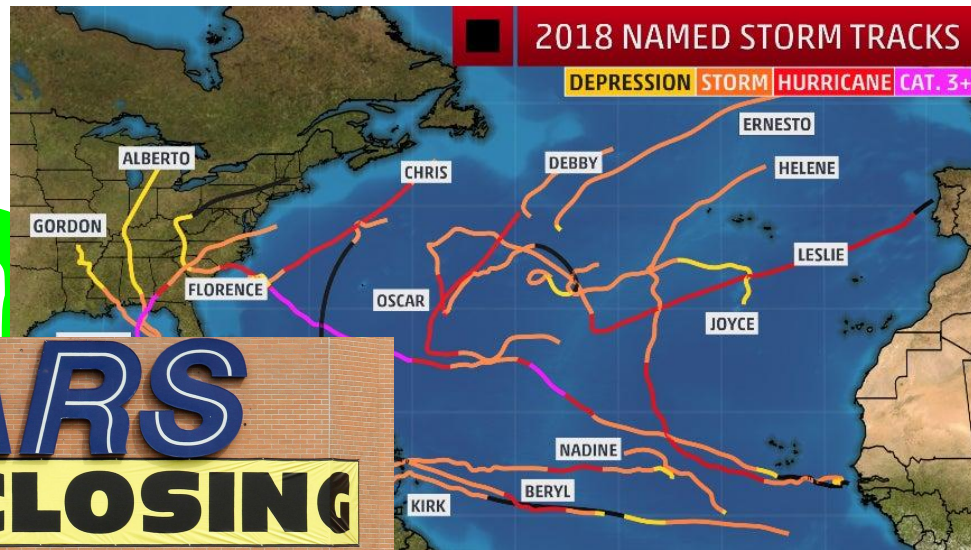
GOOD Conference at Harvard University

March 18, 2025 at 14:30-14:55

# Overview/Goals

- Quick history of OOD at Princeton
- User & cluster adoption
- What apps do we run
- Vis nodes/systemd
- Job monitoring via jobstats
- Filesystems/Quotas
- Cloud bursting
- Customizations/ERB
- Job startup and other trouble

# Quick History



# Quick History - Courses/Beginners

```
$ export XDG_RUNTIME_DIR=""
$ module load anaconda3/2024.10
$ jupyter-notebook --no-browser --port=8889 --ip=0.0.0.0
# or
$ jupyter-lab --no-browser --port=8889 --ip=0.0.0.0
# note the last line of the output which will be something like
http://127.0.0.1:8889/?token=61f8a2aa8ad5e469d14d6a1f59baac05a8d9577916bd7eb0
# leave the session running
```

If you are looking to use a custom Conda environment then see "Custom Conda Environment" above.

Next, start a second terminal session on your local machine (e.g., laptop) and setup the `tunnel` as follows:

```
$ ssh -N -f -L 8889:tiger-h26c2n22:8889 <YourNetID>@tiger.princeton.edu
```

In the command above, be sure to replace tiger-h26c2n22 with the hostname of the node that salloc assigned to you. Note that we selected the Linux port 8889 to connect to the notebook. If you don't specify the port, it will default to port 8888 but sometimes this port can be already in use either on the remote machine or the local one (i.e., your laptop). If the port you selected is unavailable, you will get an error message, in which case you should just pick another one. It is best to keep it greater than 1024. Consider starting with 8888 and increment by 1 if it fails, e.g., try 8888, 8889, 8890 and so on. If you are running on a different port then substitute your port number for 8889.

(You can also try the `get_free_port` command on our clusters)

Lastly, open a web browser and **copy and paste** the URL from the previous output:

```
http://127.0.0.1:8889/?token=61f8a2aa8ad5e469d14d6a1f59baac05a8d9577916bd7eb0
```



# Quick History - Cluster Adoption

MyAdroit - Adroit cluster

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

📁 </> ▾ ? ▾ 👤

MyDella - Della cluster

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

📁 ? ▾ 👤 ➡

Stellar cluster

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

📁 </> ▾ ? ▾ 👤 ➡

Tiger3 cluster

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

📁 </> ▾ ? ▾ 👤 ➡

Verde cluster

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

📁 </> ▾ ? ▾ 👤 ➡



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Welcome to MyTiger - web interface to Tiger3 cluster

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**OPEN**  **OnDemand**

OnDemand version: 4.0.1

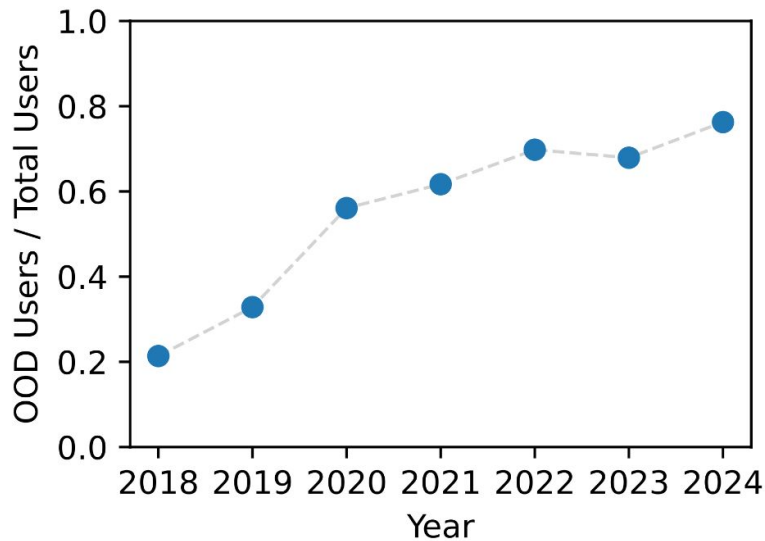
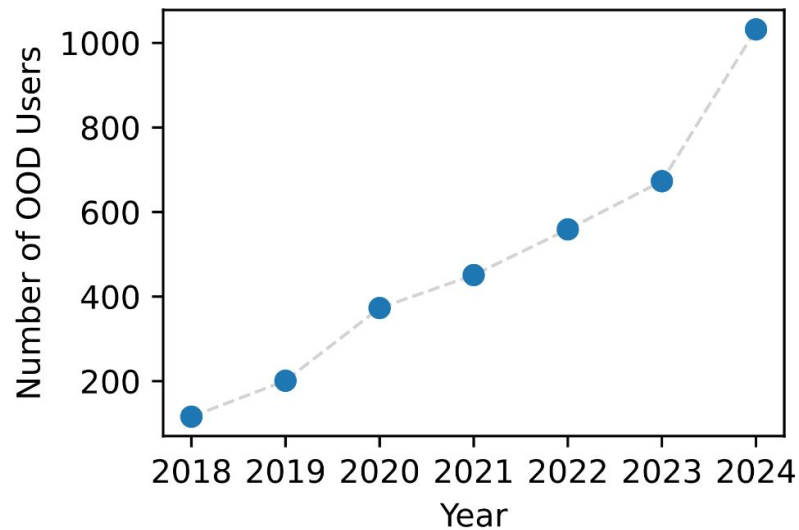


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**OPEN**  **OnDemand**

OnDemand version: 4.0.1

**OPEN**  **OnDemand**

# User Adoption Over the Years



# Applications

Desktops

Desktop (Visualization Node)

GUIs

MATLAB

Mathematica

Vivado for ECE206

XStata

Servers

CodeServer on Vis Node

Jupyter for Classes

Jupyter for Classes on Adroit Vis

Jupyter

Jupyter on Adroit Vis

RStudio Server

RStudio Server on Adroit Vis

## Jupyter for Classes

This app will launch a [Jupyter](#) server for classwork using [Python](#) on the [Adroit cluster](#).

### Class

CEE306

Please select version of python/anaconda for your class

### Partition

class

- **class** - Reserved for class use, please use this one.
- **cloud** - Cloud/AWS GPU nodes, available for select courses
- **gpu** - Allocate one GPU, no more than 3 hours.

### Anaconda3 version used for starting up jupyter interface

2024.6

2024.6

2024.2

2024.10

2023.9

2023.3

2022.5

2022.10

2021.5

2021.11

2020.7

2020.2

2020.11

Geoclim 2024.10

custom

# Visualization Nodes/Systemd

- Nodes dedicated for visualization and data processing (VNC/GPU/Python...)
- Cgroup limits on memory and CPU share
- Direct SSH logins, anything goes
- OOD Linux and Slurm schedulers bad match

```
[plazonic@adroit-vis ~]$ systemctl --user list-units --state=running 'ondemand-*'  
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION  
ondemand-TbCdhr6P0n.service loaded active running sys/dashboard/sys/bc_desktop/adroit-vis
```

```
LOAD    = Reflects whether the unit definition was properly loaded.  
ACTIVE  = The high-level unit activation state, i.e. generalization of SUB.  
SUB      = The low-level unit activation state, values depend on unit type.
```

- OOD Systemd scheduler (merged in Feb 2022)
- Same Cgroup limits - integrates well with interactive/non OOD use

# Visualization Nodes/Systemd

## Desktop (Visualization Node)

VNC based remote desktop on Adroit Visualization node  
For light interactive work that might require internet access.

Number of hours

1

☐ I would like to receive an email when the session starts

Launch

\* The Desktop (Visualization Node) session data for this session can be accessed under the [data root directory](#).

Also Jupyter, Rstudio &  
other apps

Desktop (Visualization Node) (ondemand-TbCdhR6POn@adroit-vis)

1 node | 1 core | Running

Host: >\_ adroit-vis.princeton.edu

Delete

Created at: 2025-03-14 10:09:52 EDT

Time Used: 56 minutes

Session ID: 8e455c35-b2d0-40a7-b533-2dd2b8c8e141

Problems with this session? [Submit support ticket](#)

Compression



0 (low) to 9 (high)

Image Quality

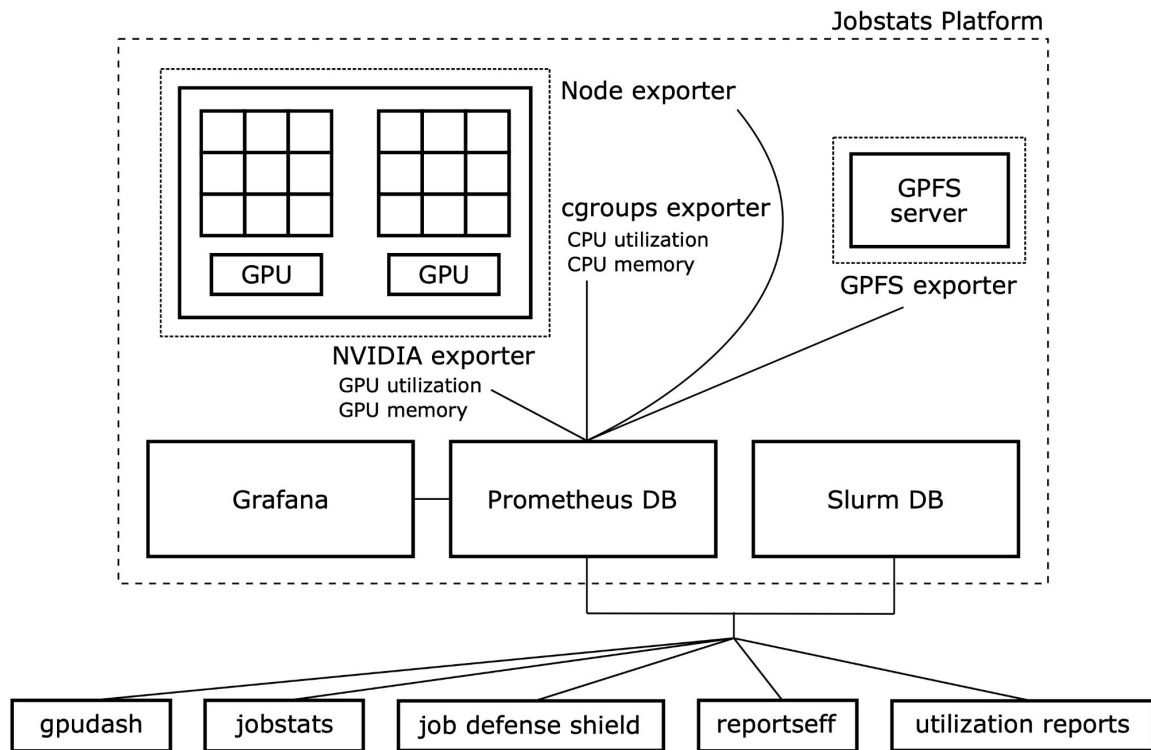


0 (low) to 9 (high)

Launch Desktop (Visualization Node)

View Only (Share-able Link)

# Jobstats Job Monitoring Platform



# Jobstats Job Monitoring Platform

- `jobstats` command shows CPU/GPU utilization and CPU/GPU memory usage
- **Automatically cancel GPU jobs at 0% utilization**
- Identify and/or send email alerts to users for 12 instances of underutilization
- Used by tens of institution throughout the world (Brown, Princeton, Yale, ...)
- Jobstats integrates with OOD

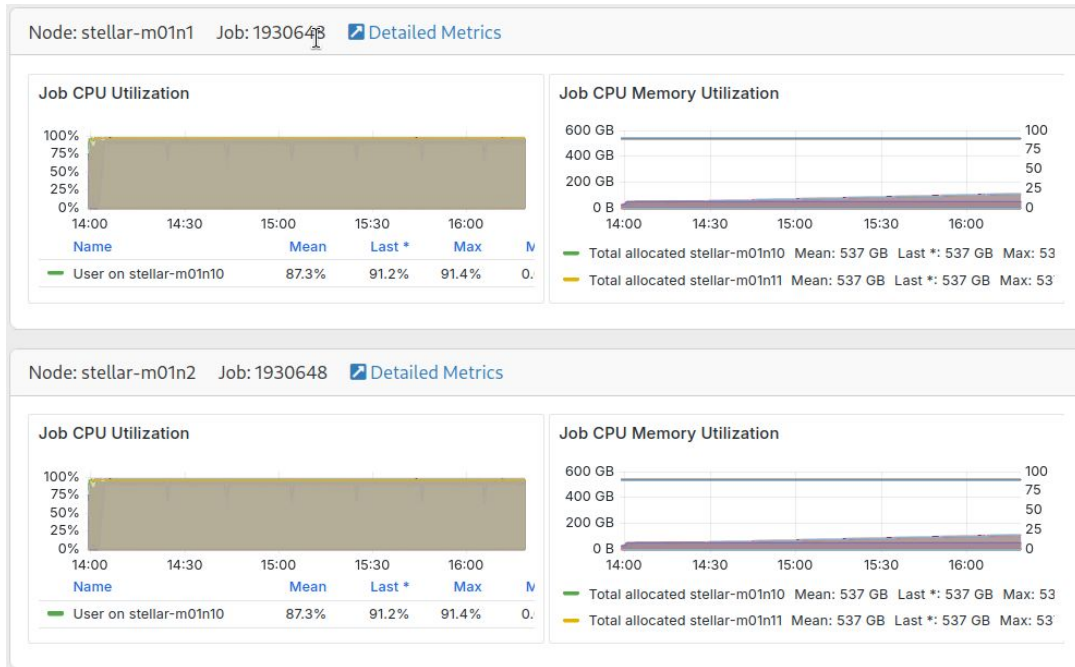
## Integration of Open OnDemand with the Jobstats Job Monitoring Platform

Wednesday, March 19, 2025 at 17:20-17:30

[github.com/PrincetonUniversity/jobstats](https://github.com/PrincetonUniversity/jobstats)

# Jobstats Job Monitoring Platform

```
---
v2:
  metadata:
    title: Super Cluster
    ....
  custom:
    grafana:
      host: https://ood.server/grafana
      orgId: 1
      dashboard:
        name: ondemand-clusters
        uid: ZAlk3fG9a
      panels:
        cpu: 7
        memory: 9
      labels:
        cluster: cluster
        host: node
        jobid: JobID
```



# Jobstats Job Monitoring Platform - Completed Jobs



The screenshot displays the Jobstats Job Monitoring Platform interface. At the top, there is an orange navigation bar with the following tabs: Files, Jobs, and Clusters. Below the navigation bar, on the left, is a logo for Prince Research Corporation. The main content area is titled 'Completed Jobs'. Below the title, there is a search bar with the text 'Your Jobs' and a dropdown arrow. To the right of the search bar, there are two date input fields: 'Start Date: 03 / 11 / 2025' and 'End Date: 03 / 11 / 2025', each with a calendar icon. To the right of the date fields is a blue button labeled 'Submit Query'. Below the search bar, there is a dark grey input field labeled 'Enter Username:'. The text 'Details are available by clicking on JobID number.' is displayed below the search bar.

Files ▾ Jobs ▾ Clusters ▾

🕒 Active Jobs

☰ Completed Jobs

🔧 Job Composer

## Completed Jobs

Your Jobs ▾ Start Date: 03 / 11 / 2025 📅 End Date: 03 / 11 / 2025 📅 Submit Query

Your Jobs

Enter Username:

Details are available by clicking on JobID number.

# Jobstats Job Monitoring Platform - Completed Jobs

Co

Ent

Job e

Show

Showing

Job 1928348 ("2xIC") [click for full stats]

Memory and CPU Efficiency

| Node          | Memory Used/Allocated    | Memory Per Core | CPU Time Used/Allocated | CPU Efficiency | Cores |
|---------------|--------------------------|-----------------|-------------------------|----------------|-------|
| stellar-m01g2 | 4.5 GB / 14.6 GB (30.5%) | 2.2 GB / 7.3 GB | 3-23:48:19 / 4-00:10:06 | 99.6%          | 2     |
| Total:        | 4.5 GB / 14.6 GB (30.5%) |                 | 3-23:48:19 / 4-00:10:06 | 99.6%          | 2     |

GPU Efficiency

| Node          | GPU      | GPU Max Memory Used/Total | Average GPU Utilization |
|---------------|----------|---------------------------|-------------------------|
| stellar-m01g2 | #0       | 11.4 GB / 40 GB (28.5%)   | 98.0%                   |
| stellar-m01g2 | #1       | 11 GB / 40 GB (27.5%)     | 98.0%                   |
| Total:        | 2 GPU(s) | 22.4 GB / 80 GB (28%)     | 98.0%                   |

State

TIMEOUT

State

CANCELLED

COMPLETED

CANCELLED by 152937

COMPLETED

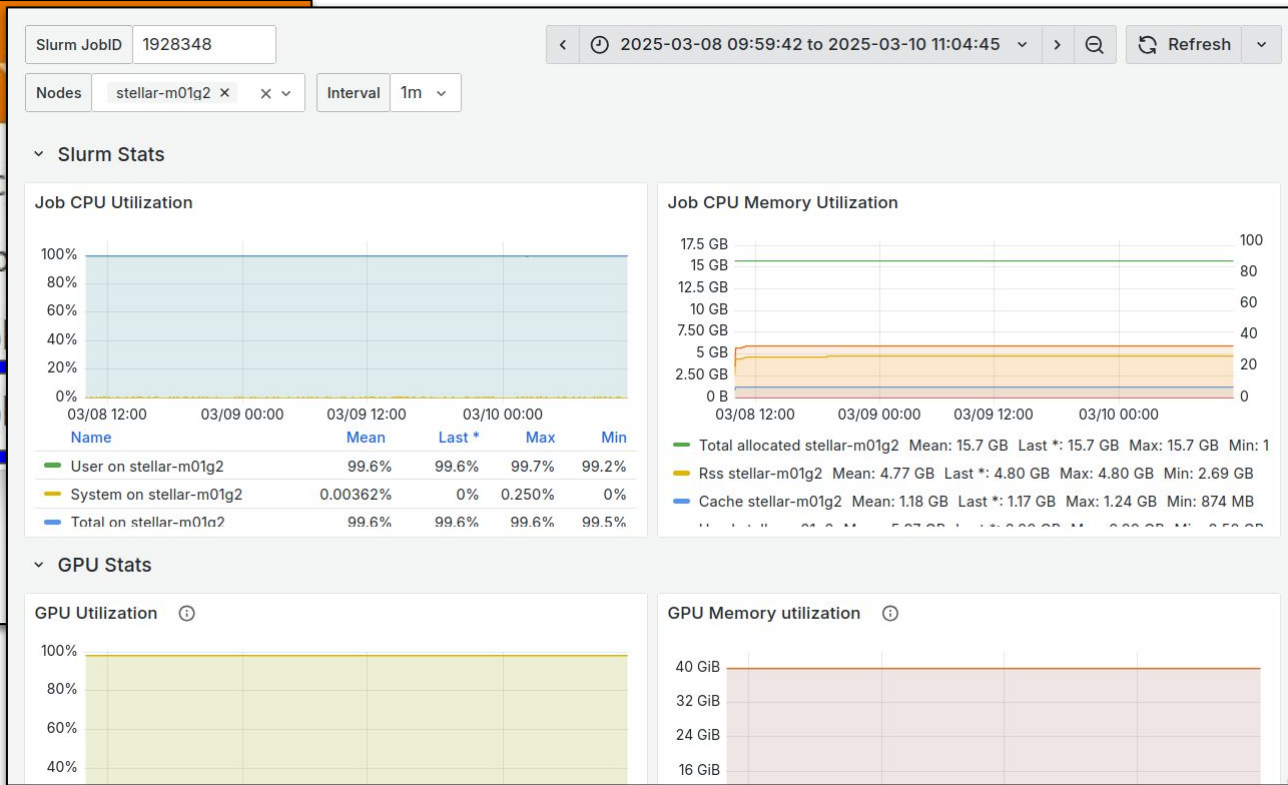
TIMEOUT

Next

Close

OPEN

# Jobstats Job Monitoring Platform - Helper



Submit Query

# Quotas/Filesystems

Files ▾ Jobs ▾ Clusters ▾

Utilities

 Quota

Home Directory

 /home/plazonic

 /scratch/gpfs/DATASETS

 /scratch/gpfs/plazonic

 /tigress/plazonic

MyDella - Della cluster / Quota report

## Quota report

For quota request increases please go to [the quota request page](#)

### Storage/size quota filesystem report for user: plazonic

| Filesystem                          | Mount                  | Used        | Limit       | MaxLim | Comment |
|-------------------------------------|------------------------|-------------|-------------|--------|---------|
| Della home                          | /home                  | 37.5GB      | 48.8GB      | 50GB   |         |
| Della scratch GPFS                  | /scratch/gpfs          | 2.9GB       | 500GB       | 512GB  |         |
| Projects GPFS                       | /tigress               | 2.3GB       | 500GB       | 512GB  |         |
| Fileset/project space               | Mount                  | Used by ALL | By plazonic | MaxLim | Comment |
| Della scratch GPFS fileset DATASETS | /scratch/gpfs/DATASETS | 21.9TB      | 1.1TB       | 25TB   |         |
| Projects GPFS fileset CCALLAN       | /projects/CCALLAN      | 2.8TB       | 4.6GB       | 10TB   |         |
| Projects GPFS fileset CEE472        | /projects/CEE472       | 9.8TB       | 578MB       | 15TB   |         |

# Bursting to the Cloud for GPUs for Classes

## Jupyter for Classes

This app will launch a [Jupyter](#) server for classwork using [Python](#) on the [Adroit cluster](#).

Class

COS324

Please select version of python/anaconda for your class

Partition

cloud

- **class** - Reserved for class use, please use this one.
- **cloud** - Cloud/AWS GPU nodes, available for select courses

- Pre-approved classes only
- Via OOD only
- VPC + direct link allows Slurm/NFS to AWS
- High latency/cloud copy of envs
- KISS

```
[plazonic@adroit5 ~]$ squeue -p cloud
```

| JOBID   | PARTITION | NAME     | USER   | ST | TIME    | NODES | ODELIST(REASON) |
|---------|-----------|----------|--------|----|---------|-------|-----------------|
| 2351525 | cloud     | sys/dash | dl4247 | R  | 2:05:32 | 1     | adroit-cloud7   |
| 2351542 | cloud     | sys/dash | ks6069 | R  | 56:02   | 1     | adroit-cloud9   |
| 2351539 | cloud     | sys/dash | aw3483 | R  | 1:02:32 | 1     | adroit-cloud8   |
| 2351519 | cloud     | sys/dash | er4138 | R  | 2:41:32 | 1     | adroit-cloud4   |

# ERB is Your Friend

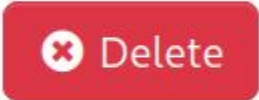
```
cluster:
  widget: "select"
  label: "Visualization node"
  options:
<%
  possibles = {
    'tiger-arrk' => 'Kingsbury/Rosen Shared Login server',
    'tiger3-sumire' => 'Astro Login server'
  }
  defnode = nil
  possibles.each do |serv, desc|
    if File.readable?("/etc/ood/config/clusters.d/#{serv}.yaml")
      defnode = serv
%>
    - ["<%= desc %>", "<%= serv %>"]
  <%
    end
  end
%>
  - ["Tiger Vis", "tiger3-vis"]
<%
  unless defnode.nil?
%>
    value: "<%= defnode %>"
  <%
    end
%>
```

```
<% envs = Dir.home + "/.conda/envs"
if Dir.exist?(envs)
  xs = Dir.glob("#{envs}/*")
  xs.sort.each do |onex|
    if Dir.exist?(onex+"/bin") &&
      File.exist?(onex+"/bin/jupyter") &&
      (File.exist?(onex+"/bin/jupyter-lab") ||
      File.exist?(onex+"/bin/jupyter-notebook"))
      x = File.basename(onex)
%>
    - ["Use your conda env <%= x %>", "<%= onex %> "]
  <%
    end
  end
end %>
```

```
<%
  username = Etc.getpwuid().name
  stdout_str, exit_code =
  Open3.capture2("/usr/bin/sacctmgr -Pn show assoc where
  user=#{username} cluster=della format=account")
  accounts = stdout_str.split("\n")
  default_account, exit_code =
  Open3.capture2("/usr/bin/sacctmgr -Pn show user #{username}
  format=DefaultAccount")
  -%>
```

# Job Startup and Other Trouble

- Job startup - significant portion of OOD related tickets
- Why is it not starting?

|                                                                   |                                                                                     |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Jupyter (62926826)                                                | Queued                                                                              |
| Created at: 2025-03-16 13:00:45 EDT                               |  |
| Time Requested: 1 hour                                            |                                                                                     |
| Session ID: <a href="#">6b00b1d5-47a0-47d4-bbe3-84ce9812ad8c</a>  |                                                                                     |
| Problems with this session? <a href="#">Submit support ticket</a> |                                                                                     |

# Job Startup and Other Trouble

- Jupyter complex startup

---

Show/hide startup details

- Starting main script (2025-03-14 15:48:42 EDT)
  - Creating kernel for /home/plazonic/.conda/envs/widg-env/ (2025-03-14 15:48:43 EDT)...done (2025-03-14 15:48:45 EDT)
  - Starting jupyter lab (2025-03-14 15:48:45 EDT)...discovered Jupyter server listening on port 59460 (2025-03-14 15:48:47 EDT)
- 

 Connect to Jupyter

# Job Startup and Other Trouble

- Why can't I connect - startup issues (e.g. RStudio failing to load large sessions)
- Why did it fail - out of RAM (RStudio/Jupyter) and other crashes
- Authentication errors (looking at your matlab-proxy)
- Systemd scheduler sensitive to shell setup
- File manager and large file uploads (silent fail on > 10GB)
- Over quota in /home breaks OOD completely - safe mode?